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Gaze Inputs for Targeting: The Eyes Have It, Not With a Cursor

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ABSTRACT

Can eye tracking enable VR users to target and select elements at par or better than controller or head-based targeting? We explored visual feedback methods (none, cursor, outline and resize) for gaze targeting with a button press for selection, and an ecologically valid representation of commercially available user interfaces with a body-locked, grid-based design. Forty participants interacted with a 5x5 square element grid, and elements subtended either 3-, 6- or 9-degrees of visual angle. If the participant looked out of the grid boundary, on button press, we chose to select the last targeted element, but no other algorithms to enhance performance were employed. We also assessed signal quality requirements with a fixed offset 1.5-degree accuracy degradation. Participants completed 36 blocks and in each, targeted and selected 15 successive elements. We found that gaze targeting, with appropriate feedback methods and a button press, can perform at par or better than the controller in cases intended to replicate targeting and selecting in world- or body-locked paradigms in AR/VR. We anticipate that with design improvements or algorithmic mitigations that this can improve significantly.

KEYWORDS

Eye tracking; user experience; input devices; 3D user interaction; human factors and ergonomics; gaze targeting

1. Introduction

Users navigate through millions of AR/VR devices by targeting and selecting elements on their corresponding world or body-locked user interfaces (UIs) (Billinghurst et al., 1998). We explored the potential advantages of targeting using eye gaze with various feedback methods and a physical-button press for selection, through a body-locked, grid-based experiment design, which we intended to be an ecologically valid representation of many UIs currently in the market. Crucially, this study targeted users without much prior experience in VR, and did not apply advanced manipulations of gaze signal (no utilization of prediction, sticky buttons, smoothing, applications of scene context or AI) (Graupner & Pannasch, 2014; Helmert et al., 2008; MacKenzie & Zhang, 2008; Miniotas & Špakov, 2004). Comparing feedback techniques is critical to assess the value of gaze as an input in future spatial computing platforms, such as in the recent announcement of the Apple Vision Pro (www.apple.com/apple-vision-pro/, n.d.). Industry and developers are already taking advantage of gaze inputs in other mainstream devices like the Meta Quest Pro, Sony PSVR 2, and Hololens 2, and the tight correlation of eye tracking signal quality and UI design requires clear empirical guidance. Gaze-based targeting as a means of interaction has long been of interest in HCI (Ware & Mikaelian, 1986), and recent advances in technology and techniques has now made it a reality (Chatterjee et al., 2015; Sidenmark & Gellersen, 2019), yet with ongoing challenges in its implementation and our fundamental understanding of its processes.

Many AR/VR devices have operating systems where users interact with 2D panels overlaid in 3D (Turner et al., n.d.). Today, these panels have different element and grid sizes enabled primarily by controller or hand input. While ISO Fitts's Law-based studies provide a baseline (Fernandes et al., 2023; Hansen et al., 2018; Qian & Teather, 2017; Zhang & MacKenzie, 2007), they do no represent UIs present today in grid-based layouts, a layout that will likely stay prevalent. In this layout, panels with grids in virtual or mixed-reality environments tend to be body or world-locked and are independent of head motion (panels are not fixed to the display (Ghasemi et al., 2021)). Therefore, we focused here on a grid-based paradigm and kept UI visual angles constant by keeping the panel we utilized at a fixed Euclidian distance from the user's head but not rotationally locked to it.

The quality of gaze as an input modality is highly dependent on eye tracking accuracy and the target size used (Feit et al., 2017). What are reasonable gaze targetable area sizes that enable effective interactions today? Using small target sizes with poor eye tracking accuracy will result in significant miss rates (Feit et al., 2017), creating a relatively frustrating experience. Still, we can bring these miss values down significantly by setting target sizes proportionally

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larger than determined gaze accuracy values, as demonstrated in modeling work (Schuetz et al., 2020). In other words, if we test gaze on larger but still feasibly sized targets within a constrained field of view (FOV), relative to gaze accuracy values measured, we might see less frustration in gaze-driven inputs. For example, with square elements at 6 degrees of visual angle per side, we could fit in $10 \times 10 = 100$ selectable elements within a 60×60 viewing angle. Furthermore, even if we set visible button sizes to be small, this doesn't limit designers from using a much larger collider (invisible interactable area) around the selectable item. Item or collider size is but one potential mitigation for eye tracking error.

Through visual feedback, a system can communicate when and which element it recognizes a user targeting (Sibert & Jacob, 2000). Different types of visual feedback produce varied effects, however. The standard gaze cursor can lead to cursor chase effects (Jacob, 1993; Lankford, 2000), where the user's eyes are constantly drawn to a cursor offset from where the user is looking, which could be an unpleasant experience. On the other hand, outlining or resizing a target can communicate targeting without the user unnaturally adjusting their eye motion (Ashmore et al., 2005; MacKenzie & Zhang, 2008; Majaranta et al., 2004; Miniotas et al., 2004; Špakov & Miniotas, 2005). This is because outlining or resizing typically happens around the intended targeted element and can be done to communicate that a user is targeting an element without capturing their attention to a particular point (Choi et al., 2020; Papelis et al., n.d.). However, this communication comes at the cost of additional selection time, as the user perceives and reacts to the feedback (Helmert et al., 2008). The interplay between different forms of visual feedback, item size, and gaze accuracy are all in need of an integrative understanding to enable gaze to be as ubiquitous as a controller for interactions in VR and AR.

Classic studies exploring the combination of gaze and pinch for targeting, selection, and interaction, have noted the limitations of cursors and instead used highlighting to provide visual feedback (Pfeuffer et al., 2017). For example, some participants found highlighting to be less distracting than other forms of feedback (Zhang et al., 2017), yet inherent in that conclusion is that there is some degree of distraction caused by the experience of the eyes causing a visible change in the stimulus. These studies did not provide an analysis of the effectiveness of these feedback approaches as a function of eye tracking accuracy; therefore the relation between the quality of the eye tracking and the forms of feedback used for input and interaction have not been explored, yet are crucial for ensuring efficient and enjoyable human-computer interaction that is based on gaze. Such an understanding would allow other potential gaze applications in the domains of AR and AI (Burlingham, Sendhilnathan, Komogortsev, et al., 2024; Burlingham, Sendhilnathan, Wu, et al., 2024) by allowing implicit gaze targeting if we find that sufficient signal quality enables gaze input without visual feedback (Sendhilnathan et al., 2024).

Putting this all together, the unique value of this study comes in combining disparate, but in some cases known elements, in a manner that makes gaze interactions feasible and effective going forward that would not work in isolation. We provide the following contributions for gaze inputs:

1. High-quality eye tracking without feedback can already outperform the controller in throughput and match it in movement time and subjective metrics. This form of eye tracking also outperforms head tracking in all measures except misses.
2. Large enough angular target sizes (6 degrees) at sub 1-degree mean gaze error, can reduce mean miss % to between 1.8% and 5%.
3. We show the relative performance of four key gaze visual feedback methods: cursor, resizing, outlining or no feedback, and we see that cursors are preferred least.
4. We demonstrate how the evaluation of visual feedback on targeting changes with different gaze accuracies and see that, while the no feedback method performs poorly and is rated poorly for smaller angular sizes with gaze degradations, this isn't the case for better eye tracking with larger elements.
5. We show that even with 1.5 degree eye tracking accuracy degradations, gaze input is comparable to the controller, and outperforms the head for larger element sizes.

For the remainder of this article, we will first explore existing works setting the foundations for world-locked grid-based gaze selection in AR/VR, and the effect of feedback methods on task performance. Next we will go into detail in our experiment design and provide the reasoning and justification of our approach. We will then dive into a set of hypotheses, and provide corresponding results. We provide an evaluation and discussion of gaze as an input in AR/VR based on our findings.

2. Related work

Before 2023, there have been conflicting reports on gaze interaction performance. Interaction performance was evaluated across different studies, with varying levels of gaze accuracy and in varied contexts using different types of tasks, target sizes, shapes, feedback types and placements, different environments, and selection techniques (Hansen et al., 2018; Hülsmann et al., 2011; Luro & Sundstedt, 2019; Qian & Teather, 2017; Rajanna & Hansen, 2018), but with each study reporting general conclusions about gaze interaction performance. Different contexts for evaluation however tend to produce different results, and many might not capture characteristics of typical VR usage. As we briefly described before, we attempt to bring different elements together in a particular context that demonstrates the feasibility of gaze in today's and upcoming UIs. A broader review has been recently published on gaze interaction in extended reality (Plopski et al., 2023).

Understandably, many eye tracking studies have placed targets locked onto a display or monitor (Schuetz et al., 2019; Ware & Mikaelian, 1986). Head-locked displays might provide metrics about eye tracking quality with fewer confounds, or input performance for this specific display-locked case. However, assessing gaze accuracy through head-locked UI does not capture the impact of natural head movements that go along with gaze inputs in targeting and selecting elements locked in the world or body-locked coordinates, which is essential for interacting with elements presented naturally in VR. In contrast, one of the contributions of our article is to focus on a particular body-locked context that better represents a majority of UI in today's most common VR devices. To our knowledge, gaze targeting and selection have not been deeply evaluated in a world or body-locked grid-based selection context for VR applications, and this has also been of interest for AR applications (Blattgerste et al., 2018).

Moreover, some ISO 9241-9 (Zhang & MacKenzie, 2007) standard-based studies exist in which targets are world-locked, but the ISO-based layouts are not representative of typical element layouts. In some scenarios, users are shown a single selectable element at a time (Fernandes et al., 2023), which, while similar, does not completely reflect user interaction found on head-independent grids, where users know where and what shape and size each element is (Agustin et al., 2009). Other studies tend to have game-based target placements (such as in shooters), which are important use cases complementary to the standard OS/application panel layout we will explore (Isokoski & Martin, 2006). In the past, there have been concerns such as "eye-fatigue" (Piumsomboon et al., 2017) or "Midas Touch" (Jacob, 1995). In our study, we avoid the Midas Touch effect by only activating elements on a controller button press (Agustin et al., 2009; Fernandes et al., 2023; Qian & Teather, 2017).

Prior studies have used various eye tracking qualities and different target shapes and sizes. The signal quality and target size need to be considered in tandem because the success of targeting and selecting an element is a direct result of the combination of these two factors (Feit et al., 2017; Fernandes, Murdison, Schuetz, et al., 2024; Fernandes et al., 2023). Many past studies did not report the eye calibration statistics (Hansen et al., 2018; Qian & Teather, 2017; Sidenmark & Gellersen, 2019) necessary to evaluate gaze as an input, as different eye tracking qualities will produce different interaction performance (Heydn et al., 2019). In prior experimental work (Graupner et al., 2008), gaze selection performance improved with larger button sizes and decreased with higher eye tracker noise. Various other works used differently shaped and sized elements placed at relatively varied locations (Hansen et al., 2018; Hülsmann et al., 2011; Luro & Sundstedt, 2019; Qian & Teather, 2017; Rajanna & Hansen, 2018), but comparing the relative success or failure of gaze input is not possible in the absence of gaze calibration quality (McConkie, 1981).

Size of target elements and layout are both crucial factors to consider when assessing gaze as an input in the literature. There is certainly a range of performance seen between studies, With a button press for selection, gaze targeting has shown significant miss (false negative) rates at approximately 40% (Qian & Teather, 2017) and 17% (Fernandes et al., 2023). In Qian and Teather (2017), different target sizes were used with cursor feedback, but the eye tracker accuracy is unknown. In Fernandes et al. (2023), the target sizes used (3,4,5 degrees) are likely a smaller range than necessary when upcoming displays can afford slightly larger targets and icons. Both studies generally followed the ISO 9241-9 reciprocal selection task, where many targets were placed towards the periphery of a display. However eye trackers tend to be most accurate in the center of FOV (Feit et al., 2017; Holmqvist et al., 2012; Schuetz & Fiehler, 2022), where a grid layout would be at an advantage. Different target sizes will produce different miss percentages (Schuetz et al., 2019, 2020), and we can likely mitigate selection misses with target sizes much larger than measured gaze accuracy errors. Here we tested three sets of button sizes (3, 6, and 9 degrees) placed in a grid layout.

When comparing gaze inputs to other input methods such as head-targeting, a controller, and even the mouse in prior work, it is important to note not only the signal quality, but also visual feedback. In many cases, low eye tracking quality limited the validity of such comparisons, e.g., reporting sub-par performance of gaze relative to head-tracking (Qian & Teather, 2017) which used a cursor. For eye gaze, cursors have disadvantages, primarily the effect of cursor chase (Hatfield et al., 1996; Jacob, 1995; Majaranta & Räihä, 2002), possibly leading to a frustrating experience. A good illustration of this in more recent work is the observation that a cursor used for eye tracking was accompanied by excell jitter, making it distracting and a hinderance to performance in the task (Shen et al., 2022). More recent studies showed eye tracking and controller targeting to be comparable in performance (Fernandes et al., 2023; Hou & Chen, 2021). Still, these studies did not include any visual feedback methodologies.

Feedback methods can reduce the ambiguity involved with gaze targeting; users have a better understanding of when the system recognizes that an element is targeted. There are many studies that have evaluated feedback methods, but here we will focus on gaze targeting and button selection paradigms. Given the ubiquity of using a cursor to show the location of mouse-input, numerous studies have examined the impact of using gaze cursors of different shapes for a variety of tasks. Some benefits were reported for using a gaze cursor as visual feedback; for example one study found it improved accuracy, yet also discovered that it slowed response times (Alonso et al., 2013). In contrast, many studies found that a cursor impaired performance and also was rated as distracting by participants (Arnold & Miller, 2022; Graupner & Pannasch, 2014; Murata et al., 2012; Papelis et al., n.d.), with cursor chase and occlusion noted as problematic (Jacob, 1995). In one case, explicitly telling users not to chase the cursor seemed to eliminate this problem (Zhang et al., 2011), though whether this would be practical in a normal consumer setting is a concern.

However, another recent article illustrated the successful use of visual feedback with a cursor for gaze input (Sidenmark et al., 2023), and reported good performance for gaze input compared to other input modalities as seen in other work without a cursor (Fernandes et al., 2023) and even when targets could appear at varying depths (Fernandes, Murdison, & Proulx, 2024). Thus, there is a better need to understand whether, when, and how visual feedback should be incorporated for gaze input. Not requiring visual feedback might also provide greater opportunities to use gaze implicitly in such XR systems, rather than only explicitly (Sendhilnathan et al., 2024).

Other options for feedback include highlighting or size changes. Some prior work has used highlighting as feedback to indicate gaze targeting; however only confounded with another form of visual feedback such as having the target simultaneously shrink in size, making it difficult to draw conclusions (Majaranta et al., 2003, 2006). Yet other studies have used a form of highlighting, such as changing the element color, but without a comparison to a no-feedback condition, so that only comparisons with other forms of feedback (e.g., haptic) could be drawn, and so do not address the research questions posed here (Majaranta et al., 2016). Size changes could also provide visual feedback. As noted, some work has had targets shrink in size which also lowers the conspicuity of the object. However, increased target size is known to improve performance and capture attention (Fernandes et al., 2023; Proulx & Green, 2011; Schuetz et al., 2019, 2020), and increasing the conspicuity would aid perceptual and decision making processes that influence the throughput and total task time. Even so, some work has raised the concern that dynamically changing the size of an element might be distracting (and therefore instead invisibly increased the invisible collider size of a target in the absence of any visual feedback) (Miniotas et al., 2004). This leaves the potential of increasing size as a form of visual feedback an open question.

This review of prior work reveals a gap in our understanding of the impact of several factors that are crucial for leveraging eye tracking as an input in VR. Specifically, we have identified that no prior work assessed how targeting may not require visual feedback if the tracking accuracy is of a sufficiently good quality. Therefore in this study we examined the impact of eye tracking signal quality and target size in the context of either no visual feedback or different forms of visual feedback. We compared gaze targeting and controller to a head-targeting baseline and presented body-locked stimuli in a grid to better represent the sorts of VR scenarios that would use such an input with increased ecological validity compared to head-locked and ISO tasks.

2.1. Hypotheses

H1: *Unmodified gaze (no degradation and no feedback) will perform similarly to the controller but outperform the head in throughput, movement times and all subjective measures.* As we used a performant Tobii eye tracker we expect "unmodified" eye tracking to perform similarly to the controller in each measure above, and outperform the head. We might get a slight performance advantage because we have larger targets that are centralized as opposed to being in the fringes of the eye trackers field of sensing, which can result in smaller miss rates (Alonso et al., 2013; Feit et al., 2017), and thus shorter movement times.

H2: *Gaze with each feedback method will perform similarly to the controller but outperform the head in throughput, movement times and all subjective measures.* Feedback will slow down time to selection, as users spend time perceiving and reacting to the feedback (Arnold & Miller, 2022; Helmert et al., 2008; Papelis et al., n.d.). We also expect participants to hold off selection till they are confident that a button is targeted. However, we do not expect this to significantly reduce performance, and we still expect controller and gaze with feedback performance to be similar.

H3: *Degraded gaze methods, when compared to the controller, will have relatively poor throughput, movement times and subjective ratings for 3-degree elements, but similar values and ratings for 6- and 9-degree elements.* We intend to see the relative effect of degraded gaze, and how that would compare against the controller. In other words, if gaze accuracy was poor, would gaze still be comparable to the controller? Once again, we understand that gaze interaction performance is not just a function of eye tracking quality, but a function of both eye tracking quality and target sizes used (Feit et al., 2017).

Our eye tracker provides an expected mean gaze error of less than 1 degree. Let's assume that after a 1.5-degree degradation, we have a p75 error of around 2 degrees. Here, if we were to use 4-degree diameter circular targets, we can expect around 25% of all gaze samples to result in misses, and more with square 3-degree elements – increasing movement times, and decreasing throughput and subjective ratings, enough for the controller to outperform it.

On the other hand, for the 6- and 9-degree elements, even with degradations applied, we expect a lower miss probability (Alonso et al., 2013; Feit et al., 2017). Again, for 4-degree diameter targets, a p75 accuracy of 2 degrees, which would result in 25% misses, but fewer samples would result in misses for 6-degree square elements. In Fernandes et al. (2023), gaze had similar throughput and subjective ratings to the controller with approximately a 17% miss rate, for smaller element sizes. With these assumptions, it is unlikely that we will see gaze performing poorly relative to the controller for larger elements. We recognize that there are many other factors in play when determining if the gaze cursor intersects with a collider, primarily those of human behavior and tendencies in eye movements, but we simplified our assumptions.

H4: *Feedback will benefit 3 degree elements the most, for both degraded and undegraded gaze.* In particular we will expect the greatest benefit of feedback when the

probability of targeting an element as intended is lowest – that is for smaller elements with degradations, which have the greatest possibility of misses. In these cases, it is likely that participants won't press the trigger until they see that the item they wish to select is indicated as targeted (Choi et al., 2020). In contrast, without feedback users will likely press the trigger multiple times until the element progresses. For larger element sizes, the probability of being unable to target an element is lower.

H5: *We expect cursors to have the lowest subjective ratings, relative to other feedback methods.* We expect cursors to have the lowest rating because of cursor chase (Choi et al., 2020; Helmert et al., 2008; Papelis et al., n.d.), and users making adjustments and corrections that take time and are not natural – which can be straining to the eye, especially for repeated motions.

3. User study

3.1. Experiment design

At a high level, as shown in Figure 1, participants interacted with a 5 × 5 square grid of white elements. We changed the color of one element at a time from white to green (target element). The participant's task was to target and select 15 elements per block with a predefined input method and feedback. Each element turned white after selection, and the next element's color changed to green. After completing 15 selections, we prompted participants with four in-app survey questions which we cover in Section 4.2. Each block consists of a combination of conditions described in detail in Section, and illustrated in Figure 2. It took on average 24 minutes for participants to run through the experiment and complete all 36 blocks.

As we show in Figure 2, we evaluated the following conditions in our experiment for gaze input:

1. Eye Feedback Methods (None, Cursor, Outline, Resize)
2. Square Element Width (3, 6, 9 degrees)
3. Eye Tracking Accuracy Degradations (None, 1.5 degrees)

We also had blocks for our input method comparisons, but repeated all blocks here twice.

1. Comparison Input (Controller, Head)
2. Square Element Width (3, 6, 9 degrees)

Visualized in Figure 2, this results in 24 blocks for gaze inputs (4 Feedback Methods × 3 Element Sizes × 2 Accuracy Degradation blocks) and six blocks each for controller and head (2 Comparison Inputs × 3 Element Sizes × 2 Repetitions). For example, one of the 36 blocks could be Eye, with Cursor feedback using element sizes of 3 degrees with no degradation.

3.2. Practice round

Before the main experiment, users ran through a condensed version of the task as practice to familiarize themselves with the interaction tasks and calibration procedure. Following eye tracker calibration, participants performed six blocks of five trials each in randomized order: four Eye Tracking (No Feedback, Outline, Cursor, Resize), one Controller, and one Head block.

3.3. Participants

We evaluated 40 participants for the study. In total we recruited 44 participants external to our organization using our databases of people who have expressed interest, and all provided written, informed consent. One participant did not meet calibration requirements, and 3 did not fully complete the experiment. Unsuccessful attempts we factored to software issues, or participants running out of time. Out of the 40 participants who completed the experiment, 39 completed the survey used to provide the stats in this section (17 female; age range 25–67; average 42.1 years old); 20 reported that they had corrected vision, out of whom 12

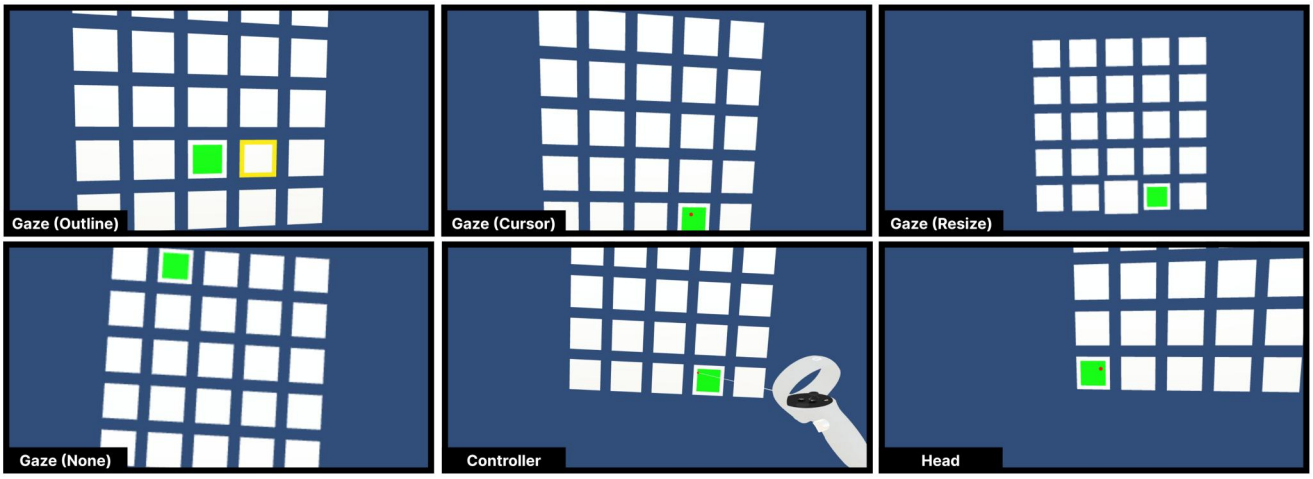


Figure 1. Participants used one of the six input methods shown above and were directed to target 15 unique elements highlighted in green, one at a time, and select each one with a controller button press.

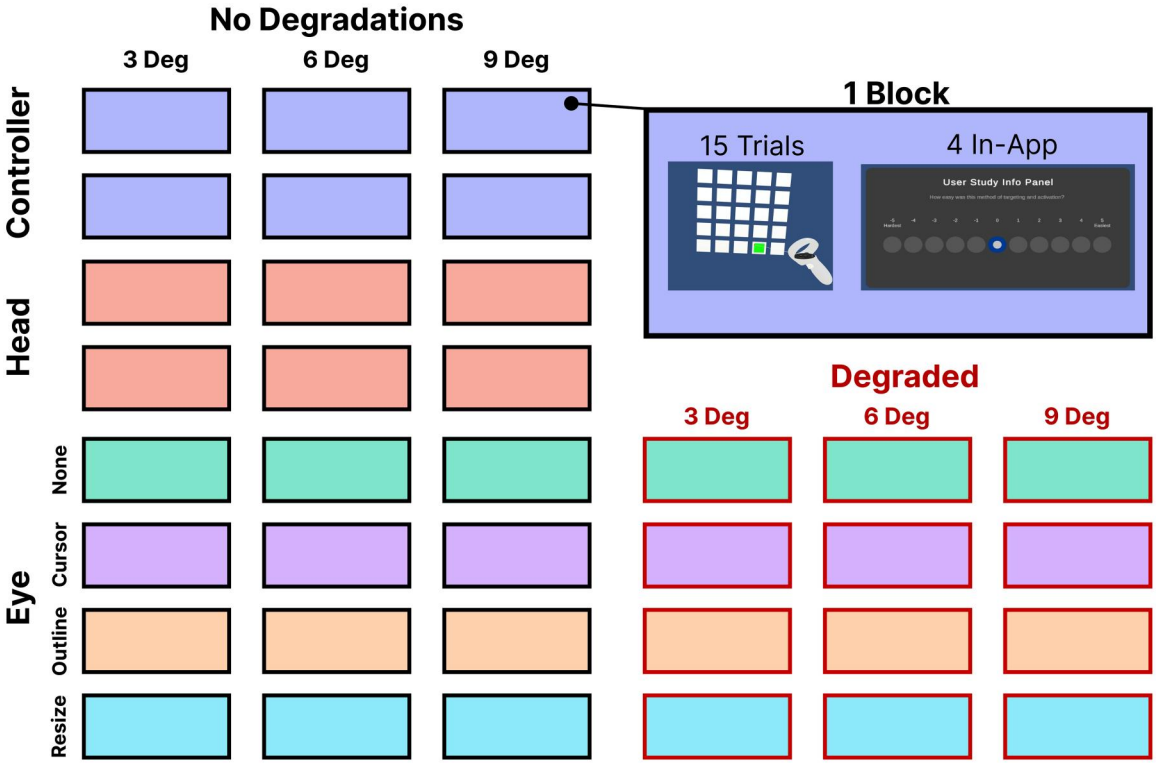


Figure 2. Participants went through 36 blocks in shuffled order, each containing 15 trials and 4 in-app questions.

wore contact lenses during the experiment, one wore glasses, and seven wore neither. We had 26 participants who indicated that they spend no time in VR per week, seven participants indicated that they spend between 0.5 and 1 hours, and 3 participants indicated they spend 1-2 hours, and three indicated they spend greater than 2.

3.4. Equipment

We used an off-the-shelf Meta Quest 2 VR headset with corresponding controllers and a display refresh rate set at 120 Hz. The display has an approximate per-eye field of view of 92×96 degrees of visual angle (deg), resolution of 1832×1920 pixels, with an approximate pixel density of 21 pixels per degree (Kim et al., 2022). We used an HMD-integrated binocular XR eye tracking platform from Tobii (Tobii AB, Sweden). This eye tracker has a sampling frequency of 240 Hz and is based on Tobii's latest generation off-axis (direct to eye) solution for VR and AR optical designs – including "pancake" lens designs common in newer VR products. We used an MSI GS66 Stealth laptop to power this experience, running an NVIDIA GeForce RTX 3070 graphics card. The experiment application was developed in Unity, and runs at an average measured frame rate of 70 frames per second, which was a bottleneck to our gaze interactions.

3.5. Eye tracking calibration

Participants started by calibrating the eye tracking system at the start of the experiment. Right after this, they were required to check their calibration quality. Calibration quality was assessed with nine targets forming a vertical cross pattern, presented singly in head-locked coordinates like the calibration, starting with one in the center and followed by four targets at 10 degrees eccentricity and four more at 15 degrees eccentricity. Participants could proceed to the rest of the experiment, assuming their p75 (75th percentile) calibration error was less than 1.5 degrees. The distribution of calibration accuracy is key to enable gaze targeting and selection, such that making a threshold just based on the median might not capture the characteristics of a skewed error distribution (Feit et al., 2017). However estimating the tail-end percentiles is challenging with small samples, making the 75th percentile a reasonable estimation to ensure that gaze targeting performance will be sufficient (Aziz et al., 2024). One of 44 participants who went through calibration was not able to make this requirement. After the first block, participants were required to check the calibration quality every ten blocks, or four times per participant. For each individual participant we averaged mean, p50, p75 and p95 error percentiles across all quality checks, followed by averaging across all 40 participants. We found mean gaze errors of 0.83 (SD = 0.36), p50 (median) of 0.67 (SD = 0.17), p75 of 0.92 (SD = 0.24), and p95 error of 2.21(SD = 2.02; all values in degrees).

3.6. Interface design and implementation

3.6.1. Body-locked design choice rationale

We considered whether to lock elements on the display to control for eye tracking errors, meaning whether elements would stay in a consistent position relative to the field of view during head movements. In studies focusing on eye tracking quality, users typically have their heads on a

chinrest and elements locked to a display (Schuetz et al., 2019; Ware & Mikaelian, 1986). This design would ultimately undermine this study's ecological validity, where a primary assumption is that users are free to rotate their heads in targeting and selecting UI elements today. There might be an assumption that a head-locked display would reduce additional noise arising from head movements or even calibration of eye gaze, yet recent work has found that gaze is more accurate when the head is free to move rather than restrained (Wei et al., 2023); this might be due to the fact that gaze is most naturally used in concert with head movements with gaze selection mostly occurring within the center of the FOV (Sidenmark & Gellersen, 2019; Stahl, 2001). Some work has examined this for input by simulating AR in VR, noting that FOV differences would need to be accounted for when interacting with UIs larger than the visible field, and indeed found that gaze targeting was superior in such scenarios (Blattgerste et al., 2018). In any case, comparing gaze to the controller or head would be challenging (if not impossible) if we locked elements to the head.

Targetable elements in many popular consumer AR/VR devices today are placed in and locked to the world in front of users – this naturally follows how everyday interactable physical objects (like books and televisions) are placed in our world, and affords natural eye, head, and body movements (Anderson et al., 2023). However, one challenge with fixed metric-sized, world-locked content, is controlling for changing perceived angular element sizes (Eggleston et al., 1996). For example, if a user cannot select small elements, they might move physically closer to the elements as a possible workaround. This action, however, compromises testing the effect of fixed angular element sizes on interaction performance. To mitigate the impact of this issue, we first asked participants to be seated and free to move their head and body, but not the chair. Through this, we limited translational movements, and would thus not affect angular elements sizes much. However, to control the possibility that users might lean in too much, we allowed the grid to move positionally based on the user's position but not rotationally, so that element angular sizes would stay fixed.

3.6.1.1. Implementation. We used a "dummy" camera that captured the position and yaw of the participant at the start of each block (Fernandes et al., 2023). The dummy camera maintained its yaw and preserved a pitch and roll of 0 throughout each block. That is, the forward vector of the dummy camera was parallel to the ground and perpendicular to gravity through all 15 trials. We did this to ensure that elements were perpendicular to the floor while maintaining their position in world coordinates across all trials per block. The most significant adjustment we propose is that the dummy camera used in this study positionally (translationally) follows the main camera.

As the grid used in this study was locked to (or a child of) the dummy camera, participants saw a world-locked grid appearing at the start of each block. This is most apparent when they rotate their head. However if they translated positionally, the grid maintained its distance relative to their head, but stayed in the same position despite their head rotation. Through this, we maintained a constant visual angle of grid elements through all 15 trials per block.

3.6.2. Grid based element layout Rationale

To test input, one standard to consider is the ISO 9241–9 reciprocal selection task (Zhang & MacKenzie, 2007). Elements are shown or highlighted one at a time in a fixed circular pattern (Soukoreff & MacKenzie, 2004). This does not wholly represent how elements are targeted and selected today, with a grid or panel of elements. Furthermore, circular element placements typically put eye-trackers at a disadvantage (Feit et al., 2017) and are not representative of typical content density – more important selectable elements tend to be placed closer to the center of a UI, than in the periphery.

Interface elements in VR today can range from 3 degrees or smaller for icons to as large as 10 degrees for images, each of which can be selected by the controller or hand in devices like the Meta Quest 2. In our study we provide targetable element sizes of 3, 6, and 9 degrees, and these choices will enable us to understand the effect of gaze-targeting on a range of element sizes. We will also see the impact of visual feedback methods on elements of various sizes. To simplify our calculations, we removed any curvature from our designs and placed elements on the same plane, with its closest distance at 1 meter away from the dummy camera.

3.6.2.1. Square element colliders. We choose to use square-shaped colliders over circular-shaped colliders because they cover the most surface in targetable areas in a UI. A simple implantation with circular colliders would have empty areas between them. We recognize that circular elements may make calculations easier to account for or standardize, though we chose the path that likely represents practical implementations of gaze-based UI in the future. Additionally, while we provide results for a square collider this does not prevent designers from using other shapes for the contained graphical element.

3.6.2.2. Implementation. As we show in Figure 3, we will differentiate a potential "element" in a grid by total collider length, visible area length, and indicator length. For 6-degree elements, we mean that the collider's length and width are both 6 degrees. We set the visible area length and width to be 75% of that size (here: 4.5 degrees). Finally, the indicator, which we highlight green, would be $4.5 \times 0.75 = 3.375$ degrees, as measured from the center of the display. Each element in the grid, for each block, has the same metric size and maintains this throughout the block. We also chose this design for more practical reasons: to enable outlining as a feedback method, where we change the color of the visible area of the element, not including the indicator itself. Additionally, the visual space in between allows elements to resize while still not overlapping with other elements.

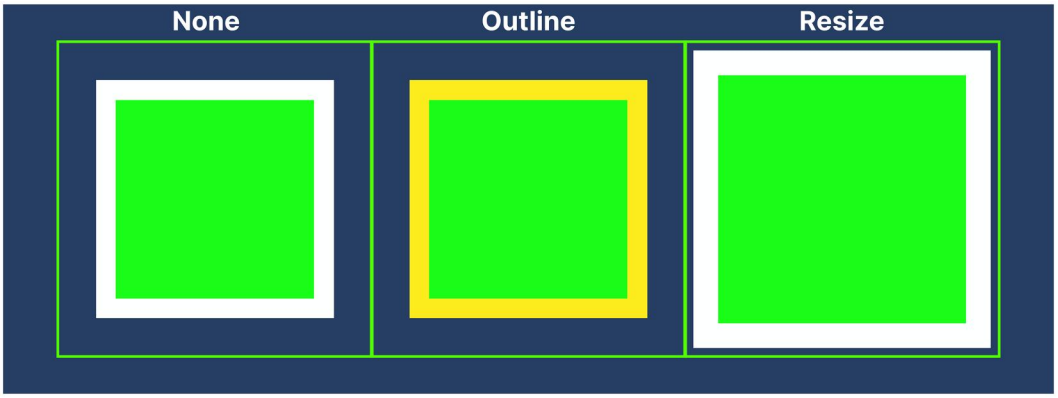


Figure 3. By default, a green indicator is set to 75% of the length of white visible area which was set to 75% of the collider length (invisible to user, but shown as a green border here). On targeting, "none" would remain unchanged, "outline" would immediately change visible area's color, not including image to yellow, and "resize" would grow visible elements to 125% its default size, but not the collider.

3.6.3. Input methods Rationale

Like in Fernandes et al. (2023), we compared gaze with controller and head tracking. Controller tracking is the current "gold standard" of input across most mainstream VR devices (Anthes et al., 2016). We chose to leverage controller tracking as the potential upper bound of input and interactions. Head tracking also serves as a potential fallback method (Sidenmark et al., 2022), therefore we captured its relative performance here for future work.

3.6.3.1. Implementation. We replicated input strategies from Fernandes et al. (2023) for controller and head where we raycast from the center of the head (main camera) or in the forward vector direction of the Meta Quest 2 controller. We use a flat red cursor on the raycast intersection point for both inputs, but use a thin rod visualizing the controller's forward direction. For the eye condition, we raycasted from the Tobii gaze origin point in the direction of the gaze direction vector provided, where the no feedback targeting method was identical to Fernandes et al. (2023).

3.6.4. Feedback methods Rationale and implementation

We chose four types of gaze feedback methods to cover a range of more precise possibilities. Targetable elements would either have (1) no feedback, (2) traditional cursor-based feedback, (3) simple adjustments to the color/texture (but not the shape of targetable element), as we do with our outline method, and finally (4) adjustments to the shape/motion of the targetable element as we do with resizing. We recognize there are many different ways of implementing each one of these branches of methods, and we attempted to keep our implementation as generic and simple as possible.

3.6.4.1. None. None, that is, no feedback targeting, works with high-quality eye tracking because users tend to know implicitly that their target of interest will get selected – what they look at gets selected on a button press. Feedback on targeting may be less valuable for gaze input that is entirely reliable. Conversely, the controller requires a user to pay attention to online feedback (typically by a visible ray or cursor (Jota et al., 2010)) and then adjust their hand to move the controller to a desired position. In our implementation, we provided no feedback except the implicit feedback that the element progressed to the next.

3.6.4.2. Cursors. Cursors have traditionally been used in gaze (Alonso et al., 2013; Arnold & Miller, 2022; Graupner & Pannasch, 2014; Murata et al., 2012; Papelis et al., n.d.; Qian & Teather, 2017) but have many disadvantages like occlusion and cursor chase. We may see corrective behavior with smaller targets (where participants unnaturally move their eyes to adjust the cursor position relative to an element) (Hatfield et al., 1996; Jacob, 1995; Majaranta & Räihä, 2002). However, cursors might be unnecessary for large elements– with an excellent eye-tracking system, this highly accurate feedback might be overkill, where a simple outline might be sufficient. Our implementation provided a small flat red cursor measuring 0.5 degrees in diameter. The cursor represents the point of collision of the gaze raycast and grid panel, and was placed right in front of the targetable elements.

3.6.4.3. Outlining. Outlining works by changing an element's color or texture (Choi et al., 2020; Majaranta et al., 2003, 2006, 2016; Papelis et al., n.d.). This, however, could change the content of a button in an undesirable manner – a designer might prefer not to change the color or texture of an image or icon. We chose not to adjust the texture or color of the image that could be potentially present on the button, and decided instead to provide a border color change. In our implementation the button is outlined yellow the moment the raycast overlaps with a particular collider, and reverts back to white immediately otherwise. To form a baseline we chose not to provide any form of fixational delay prior to outlining.

3.6.4.4. Resize. Resize animates an element (Majaranta et al., 2016; Miniotas et al., 2004). We do a simple resize of elements as a generic method to alter the form of an object. There are more specific options to choose from, such as changing the shape of an object or giving it a slight nudge and rotating it slightly. Animations that take time could affect performance. Still, methods like these should be explored but

are out of the scope for this study. We resized only the visible elements of the button and not the collider. This means that the actual area for targeting and interacting with the button remained the same but only the visual aspects perceived by the user change. Resize parameters were 125% of the original length of the element size and were returned to their original size in the same frame the raycast exited the collider.

3.6.5. Accuracy degradation rationale

Many eye trackers do not have sub-1-degree accuracy (Pastel et al., 2021; Schuetz & Fiehler, 2022; Stein et al., 2021; Wei et al., 2023). How would human performance change for less performant systems than those used in our study? Some feedback methods might be effective for high accuracy but not for low accuracy eye tracking. Simulating exact degradations is a nuanced topic, as eye trackers have unique characteristics (Holmqvist et al., 2012) and models can only approximate real tracker error (Schuetz et al., 2020). Still, we tried to keep our method as simple as possible and apply a simple linear accuracy degradation of 1.5 degrees to half of all gaze blocks as used in past work (Fernandes, Murdison, Schuetz, et al., 2024). It is important to note that participants were not informed of which blocks had gaze degradations.

3.6.5.1. Implementation. For each of the 12 degraded experiment blocks, we calculated a new gaze position, by adding a constant offset to the existing system gaze point every frame. Offsets were placed at a random rotation at the start of each block around the original gaze point, which stayed constant through all trials in the block.

Accuracy offsets were calculated in head relative coordinates in the following manner. To get a new x,y in degrees, at the same z-distance, $(P_{degraded})$, from the original Tobii gaze point $(P_{original})$, with a predetermined rotation θ, and offset magnitude *Mag*. Note that through this method, while we might be adding 1.5 degrees of error to the original gaze point, the resulting error, relative to a targetable element, might be less than 1.5 degrees. Raycasts used for target progression were shot through $(P_{degraded})$.

$$P_{degraded} = P_{original} + Vec3(\mathrm{Cos}(\theta), \mathrm{Sin}(\theta), 0) * Mag \quad (1)$$

3.6.6. Target progression

Every time the input raycast overlaps with a grid element, that grid element is temporarily stored in memory. If the raycast overlaps with a new grid element, the prior element in memory is immediately replaced with the new element. The element stored in memory is not replaced if the user looks at space outside the grid. Depending on controller trigger presses, one of the following progression criteria could occur:

3.6.6.1. Hit. If the element stored in memory is the highlighted element and the participant pressed the Meta Quest 2 controller trigger, the element progressed immediately to the subsequent element. If the input ray is outside the 5×5 grid, but the element highlighted was targeted prior, because the prior element is stored in memory, a progression happens as well.

3.6.6.2. Miss. If the element stored in memory is not the highlighted element, and the participant pressed the controller trigger, nothing would happen, and the user could attempt again. Each miss is recorded.

3.6.6.3. Timeouts. Participants could keep trying to hit the highlighted element, until it automatically advanced to the next one in the sequence after a 5-second timeout. We chose this because a longer selection would likely be a poor form of input. In the literature, movement times (from element highlighting to successful selection) often average one second or less (Fernandes et al., 2023; Schuetz et al., 2019; Sidenmark & Gellersen, 2019). Time up to five seconds was included in our movement time calculations. If a timeout occurred, a value of 5 seconds would be recorded as the movement time.

4. Measures

4.1. Objective measures

To simplify our calculations, we placed elements at the same depth, as changes in target depth impact performance (Fernandes et al., 2024). There has been a lot of debate on how to compare input and interactions when using Fitts's Law, and thus we utilize Movement Time as a key metric of performance in our study as well. Importantly, here we only utilize Fitts's equations in order to quantify relative ease of interaction for different methods using the notion of "throughput" in our study, as in past work in the literature (e.g., Fernandes et al., 2023), rather than creating an end-to-end Fitts's style experiment and analysis with the most current information-theoretical metrics for comparison with interaction methods beyond this study. To calculate throughput, we first have to calculate Movement Time (MT) and Index of Difficulty (ID).

4.1.1. Movement time (MT)

We calculated MT as either (1) the time from when an element is highlighted to when the user successfully selects that element, or (2) 5 seconds, if the element times out. We use this value in our per-target/trial throughput calculations.

4.1.2. Index of difficulty

We used the Shannon Formulation Fitts's Index of Difficulty, most frequently used in targeting and selection tasks (MacKenzie, 1989). Here we have the distance to the target (D) and target size (W).

$$ID = log_2(D/W + 1) \quad (2)$$

We choose to follow calculations provided by Fernandes et al. (2023). We calculated ID by using the dummy camera and the distances between element centers relative to that camera in degrees. For this study, we need to establish how we calculate "W" since our study used square elements, which have larger targetable areas, as opposed to circular

targets used by Fernandes et al. (2023). Note that the distance from the center of the square to the corner of each square is longer than the diameter of a circle defined by "W" in the ID equation. All measurements are in degrees relative to the dummy camera, including sizes of elements.

$$W = AverageSize = (Height + Width + Diagonal)/3 \quad (3)$$

$$TP = ID/MT \quad (4)$$

Since all elements have the same metric sizes, those further away from the center will have slightly smaller angular sizes – and these angular sizes were used to calculate W. The target size of 3, 6 or 9 degrees is only used to represent the size of a central button, placed at the center of the dummy camera, but all other targets, which as a function of having the same metric size, will have smaller angular sizes when placed on the same plane away from the camera.

One important point to note in the future, is how do we measure throughput for interfaces that are inherently predictive – that is while their visual size is smaller, their progression is based on predictive methods. Particularly in this case, a small W might not be particularly relevant – as the element might get selected with ease even though it is small in size, with the assistance of AI. In our case, our "W" might be slightly larger than the target's size for elements that are selected outside of the grid. However this brings into question another measure of performance for elements in part selected in a probabilistic manner in the future, despite having small visual/collider sizes.

4.1.3. Miss calculations

We include every time a participant pressed the button but did not successfully select an element, which resulted in a miss. A participant could miss multiple times before proceeding.

$$miss\% = \frac{\#MissTriggerPresses}{\#MissTriggerPresses + \#HitTriggerPresses} * 100perblock \quad (5)$$

4.2. In-app survey

Following each block, we adopted panel designs and three questions from existing work (Fernandes et al., 2023). We showed participants 4 separate panels, containing one question each. Questions were presented in a Likert scale format, with 11 possible responses. Default values were median values. Participants were verbally informed that desirable values were to the panel's right, and undesirable values to the left.

Q1 Ease of use: How easy was this method of targeting and activation? **A1**: -5 hardest, 5 easiest, Default value 0. All integers in between were visible.

Q2 Adoption: How much do you agree with the statement: "I am willing to use this as a method of targeting and selecting in an AR/VR device"? **A2**: -5 Strongly Disagree, 5 Strongly Agree, Default value 0. All integers in between were visible.

Q3 Fatigue: Rate your perceived exertion (all feelings of physical stress and fatigue) to target and select. **A3**: We presented 11 options with the left most element titled "Maximal Exertion" and the right most titled "No Exertion." Options in between were simply lines.

Q4 Frustration: How frustrating was it to complete the selection task? **A4**: -5 Very Frustrating, 5 Not Frustrating, Default value 0. All integers in between were visible.

4.3. User preferences, experience and feedback

Each participant also had to fill out a questionnaire through Google Forms for demographic information and questions regarding their experience. This was used to assess qualitatively the subjective experience of the users on their preferred method of input and freeform observations, in addition to their prior gaming or VR experiences.

5. Statistical assumptions

A nuanced analysis of particular comparisons of input type, feedback method, and target size will provide the most value to advancing the field given the past literature examining these variables independently (Feit et al., 2017; Fernandes et al., 2023). Therefore, we focused on hypothesis-driven groupings to compare the effects of the independent variables on the dependent measures. This approach reduces familywise error and the potential for spurious findings (Klugkist et al., 2014), yet due to the number of comparisons we have used the Bonferroni-Holm correction (Giacalone et al., 2018). Pairwise comparisons can be susceptible to violations of normality, so if a nonparametric test was required due to the difference between conditions violating the assumption of normality then a Wilcoxon signed-rank test was also used. Most pairwise comparisons did not violate the assumption of normality, and those that did (e.g., Throughput for outline versus resize conditions with 1.5 deg degradation; Movement Times including the Head condition; the majority of conditions for Misses that had no degradations), had the same inferential result for the Wilcoxon. All tests for significance were made at the $\alpha = 0.05$ level.

5.1. Timeouts

As we explained earlier, timeouts were included in throughput and movement time statistics. On a timeout, movement time for the particular trial was set to 5s. 494 out of 21600 (2.287%) potential trials for all participants resulted in timeouts. Broken down, these include 23 of 3600 (0.639%) controller trials, 2 of 3600 (0.028%) head trials, 37 of 7200 (0.514%) undegraded gaze trials, and 432 out of 7200 (6%) degraded gaze trials. For throughput and movement time, we removed the first trials of each block from our analysis. To analyze both **H1** and **H2**, we ran t-tests for Throughput, Movement Time, and the four subjective questions corrected for multiple comparisons.

6. Results

6.1. H1 gaze without feedback performance

If we inspect the performance graphs and tables (Figure 4, Table 1), the no feedback option (green), with no degradations, appears to perform on par or better than the controller in throughput and movement time, and we see similar trends visually when comparing subjective metrics (Figure 5, Table 2) between controller and gaze. Comparing unmodified eye tracking (no degradation and feedback methods) to the controllers, we see that in terms of throughput, gaze outperforms controllers, refuting the null hypothesis ($p < 0.017$). We see that in terms of movement time gaze performs similarly to the controller. In terms of subjective metrics, we see consistently that eye tracking is rated similarly across all four questions to the controller (all $p > 0.05$).

As expected, and shown visually in the graphs (comparing green in the no degradation column and red bars), gaze outperforms the head in terms of higher throughput ($p < 0.001$) and lower movement times ($p < 0.001$). Similarly unmodified gaze is rated higher than head in all four in-app questions ($p \leq 0.003$).

6.2. H2 gaze with feedback performance

We see no significant differences in throughput between each gaze method (cursor, outline or resize), when individually compared to the controller, as shown in the boxplots (purple, orange and light blue) in Figure 4. We do see that the cursor seems to perform the worst relative to the other three gaze feedback methods in each of the no-degradation charts, as well as in subjective charts (Figure 5). In terms of movement time, we see that only the gaze cursor performs poorly relative to the controller ($p < 0.001$). This pattern is replicated for every single subjective question, where only gaze with the cursor receives lower ratings than the controller ($p < .035$).

Gaze with feedback methods outperforms the head in terms of throughput ($p \leq 0.006$), however with movement time, only the cursor doesn't outperform the head ($p = 0.56$). For Q3 (Fatigue), all feedback methods are rated

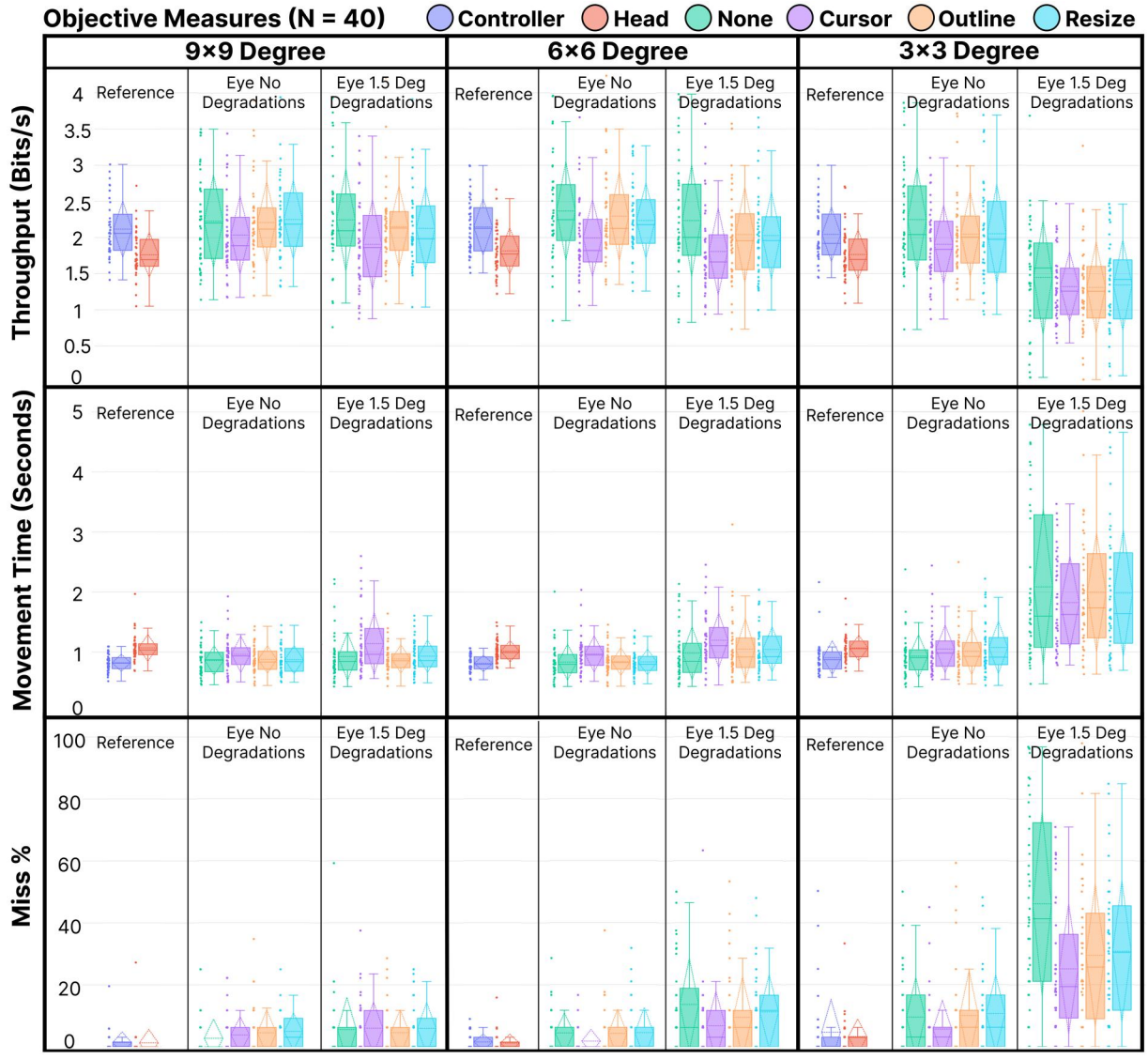


Figure 4. Objective measures, each dot is a participant's performance, by element sizes, input, and with or without degradation.

Table 1. Objective measures, by element sizes, input, and with or without degradation.

		No Degradations						1.5 Deg Accuracy Degradation					
		9 Degree		6 Degree		3 Degree		9 Degree		6 Degree		3 Degree	
		mean	stdev	mean	stdev	mean	stdev	mean	stdev	mean	stdev	mean	stdev
Throughput (Bits/s)	Controller	2.12	0.39	2.15	0.39	2.05	0.38						
	Head	1.76	0.32	1.81	0.32	1.77	0.35						
	None	2.23	0.65	2.37	0.69	2.25	0.74	2.24	0.71	2.24	0.78	1.45	0.80
	Cursor	2.04	0.55	2.00	0.54	1.91	0.55	1.91	0.63	1.81	0.61	1.32	0.49
	Outline	2.21	0.57	2.30	0.62	2.05	0.59	2.15	0.57	2.04	0.67	1.31	0.64
	Resize	2.25	0.55	2.24	0.53	2.05	0.69	2.13	0.59	2.03	0.61	1.34	0.63
MT (s)	Controller	0.82	0.14	0.81	0.14	0.90	0.28						
	Head	1.07	0.23	1.01	0.18	1.06	0.21						
	None	0.86	0.25	0.83	0.30	0.93	0.36	0.92	0.40	0.98	0.45	2.09	1.32
	Cursor	0.96	0.31	0.97	0.31	1.05	0.40	1.14	0.52	1.19	0.46	1.82	0.78
	Outline	0.87	0.23	0.84	0.23	1.02	0.40	0.88	0.24	1.05	0.51	1.99	1.00
	Resize	0.87	0.24	0.83	0.21	1.08	0.44	0.93	0.27	1.04	0.38	1.98	1.07
Miss %	Controller	1.4	3.5	1.6	2.6	4.7	10.6						
	Head	1.3	4.4	1.2	2.9	3.3	5.7						
	None	2.8	6.2	4.4	6.5	9.6	12.9	5.6	10.5	13.7	15.1	46.1	30.1
	Cursor	3.8	6.0	1.8	3.8	5.7	9.3	6.0	8.6	6.8	11.2	25.1	21.2
	Outline	4.5	7.5	4.3	8.0	10.1	14.6	4.6	7.5	9.5	13.2	29.5	23.3
	Resize	5.0	6.0	4.5	8.2	10.7	12.6	6.0	8.5	11.3	12.1	30.7	23.1

$N = 40$.

higher than the head ($p < 0.01$), but for Q1, Q2, and Q4 only the cursor is rated similarly to the head ($p \geq 0.113$).

6.3. H3: Degradations

When visually inspecting Figure 4, eye tracking with degradations performs the worst for 3 degree elements. However, this trend seems to be less apparent as we increase element sizes to 6 and 9 degrees; that is, even with degradations, performance (throughput and movement time) seems to be more in line with the controller and undegraded options, but we see this trend continue with subjective evaluations (Figure 5). We visually see the cursor seems to have the worst performance. To assess H3 we evaluated element size (3, 6, and 9 deg) and degradation of the eye gaze signal as a function of the four gaze feedback methods and the controller for each of throughput, movement time, and the four subjective questions.

6.3.1. Degree elements

With degradations applied, we see that every single degraded eye method performs poorly relative to controller in 3 degree elements for throughput and movement time ($p < 0.001$). We see similar effects in each of the four subjective questions where controller is rated higher than the degraded eye, except for one case in Q3 (Fatigue) where degraded eye (with no feedback) compared similarly to controller ($p \geq 0.071$).

6.3.2. 6 And 9 degree elements

With degradations applied, we see that all degraded gaze methods perform similarly to the controller in terms of throughput except in one instance, where Eye with Cursor (6-degree elements) performs poorly relatively ($p < 0.030$). In movement times we see that degraded eyes perform similarly to controller ($p \geq 0.181$) throughout. Finally in terms of subjective metrics, we see that (None, Resize and Outline) is rated similarly to the controller in all questions. However, this pattern is broken by cursor, where degraded cursor is rated poorly relative to the controller for Q1-Q4 with 6 deg elements, and Q2 (Adoption) and Q4 (Frustration) for 9 degree elements ($p < 0.05$).

6.4. H4: Miss percentages

When visually inspecting (Figure 4), for undegraded gaze, we see more misses for 3 degree elements as expected relative to 6 and 9 degree elements. Within that visually, we see that None seems to have more misses than the other three eye tracking methods. With degraded gaze, we see a significant number of misses for the 3 degree elements, but this number

goes down significantly as we increase element size. However for the three degree elements, we see the most number of misses for None, relative to other feedback methods, but this difference is not apparent for larger elements. We therefore examined undegraded vs degraded output containing the 4 gaze feedback methods and the 3 element sizes.

For undegraded gaze, feedback methods were not significantly different in miss percentage for any of the element sizes (3 vs 3 deg, 6 vs 6 deg, 9 vs 9 deg); neither of Cursor, Outline or Resize methods reduced miss percentages ($p \geq 0.63$), when compared to None.

However, for degraded gaze, and specifically the 3-degree elements, feedback significantly reduces errors for each of the three feedback methods, relative to "None" feedback method ($p < 0.001$). However, we continue to see the trend where feedback type doesn't have an effect on miss percentage for 6- and 9-degree elements.

6.5. H5:Subjective ratings, feedback methods

With a quick visual inspection of Figure 5 and Table 2, we do see that among gaze feedback methods, that cursor tends to be the worst rated in all questions. We evaluated undegraded vs degraded output containing the 4 gaze feedback methods, on the 4 subjective questions. We see that the cursor significantly receives negative ratings ($p < 0.013$) relative

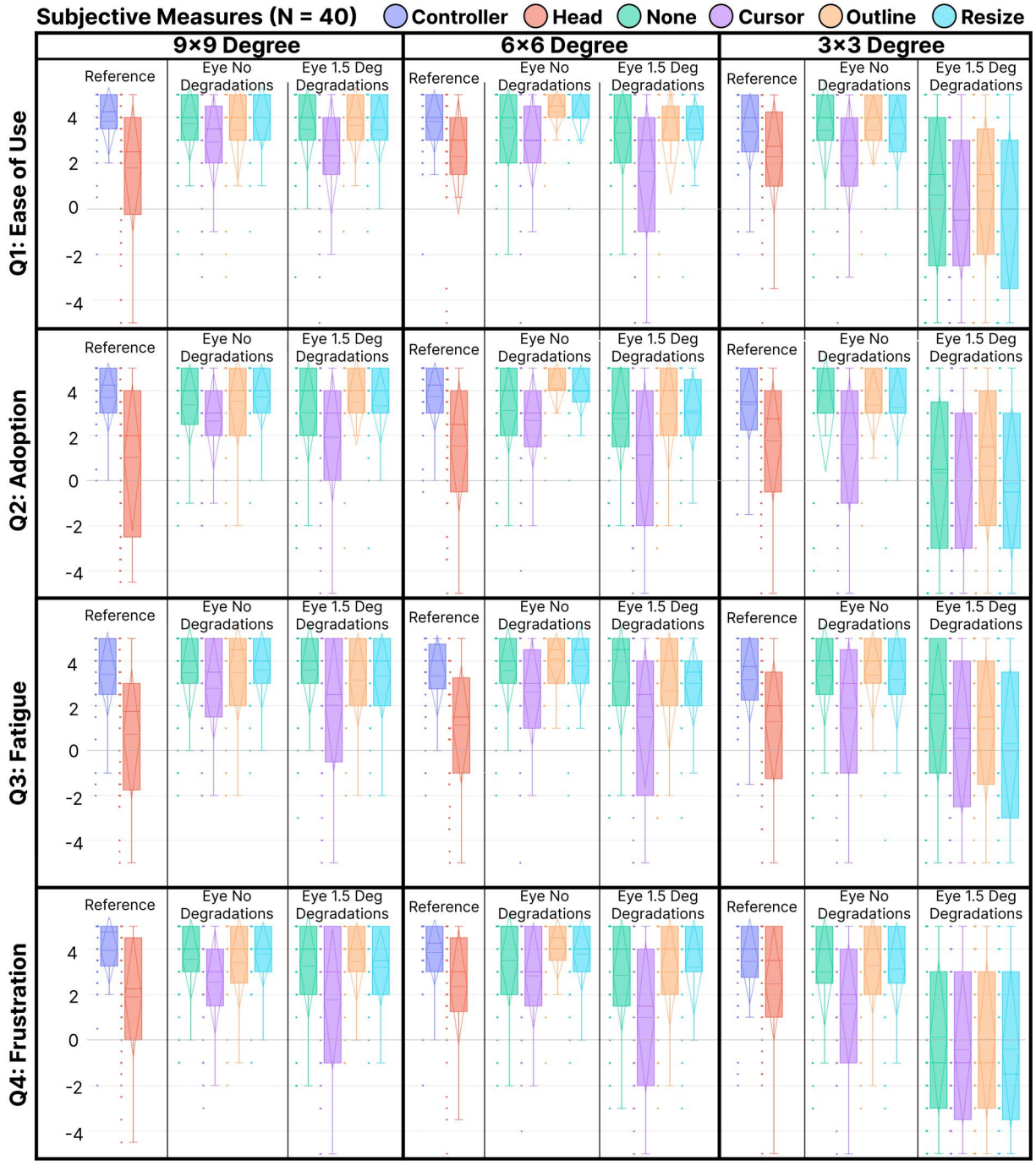


Figure 5. Subjective measures, each dot is a participant's evaluation, by element sizes, input, and with or without degradation.

Table 2. Subjective measures, by element sizes, input, and with or without degradation.

		No Degradations						1.5 Deg Accuracy Degradation					
		9 Degree		6 Degree		3 Degree		9 Degree		6 Degree		3 Degree	
		mean	stdev	mean	stdev	mean	stdev	mean	stdev	mean	stdev	mean	stdev
Q1 Ease of Use	Controller	3.84	1.51	3.83	1.40	3.39	1.72						
	Head	1.79	2.84	2.29	2.56	2.29	2.53						
	None	3.73	1.66	3.55	1.74	3.45	2.24	3.48	1.85	3.33	1.90	0.60	3.56
	Cursor	2.93	2.08	2.98	2.11	2.33	2.57	2.33	2.65	1.65	2.99	-0.05	3.14
	Outline	3.45	2.24	4.25	0.95	3.45	1.58	3.65	1.73	2.98	2.35	0.78	3.34
	Resize	4.00	1.43	3.95	1.13	3.30	1.65	3.45	1.48	3.33	1.64	-0.03	3.55
Q2 Adoption	Controller	3.70	1.56	3.74	1.53	3.39	1.61						
	Head	1.04	3.31	1.54	2.98	1.78	2.72						
	None	3.38	1.84	3.13	1.95	3.00	2.63	3.03	2.27	2.75	2.38	0.35	3.52
	Cursor	2.65	2.17	2.68	2.15	1.60	3.03	1.93	2.93	1.15	3.32	0.00	3.31
	Outline	3.53	2.00	4.10	1.15	3.35	1.63	3.50	1.95	2.98	2.50	0.65	3.43
	Resize	3.73	1.47	3.98	1.17	3.25	1.77	3.33	1.69	3.10	1.81	-0.13	3.47
Q3 Fatigue	Controller	3.40	1.90	3.34	1.83	3.16	1.95						
	Head	0.74	2.93	1.14	2.70	1.29	2.76						
	None	3.48	2.09	3.58	1.95	3.35	2.30	3.60	2.02	3.08	2.51	1.68	3.16
	Cursor	2.78	2.29	2.63	2.44	1.90	3.06	2.03	3.00	1.50	3.19	0.55	3.44
	Outline	3.48	2.20	4.08	1.16	3.38	1.97	3.15	2.13	2.68	2.63	0.98	3.32
	Resize	3.60	1.75	3.78	1.80	3.18	2.27	3.33	1.93	3.00	1.96	0.33	3.45
Q4 Frustration	Controller	3.94	1.49	3.85	1.46	3.45	1.91						
	Head	1.90	2.68	2.35	2.58	2.46	2.55						
	None	3.55	1.78	3.50	2.04	3.00	2.42	3.25	2.32	2.85	2.51	0.13	3.44
	Cursor	2.55	2.33	2.83	2.37	1.60	2.70	1.78	3.15	1.00	3.37	-0.43	3.35
	Outline	3.40	2.05	4.03	1.37	3.28	2.16	3.45	1.74	3.00	2.38	0.03	3.53
	Resize	3.78	1.42	3.78	1.61	3.13	2.09	3.20	1.71	3.20	2.11	-0.40	3.56

$N = 40$.

to other feedback methods (None, Outline, Resize) for all questions in both degraded and undegraded methods.

7. Discussion

Our study results show that gaze inputs without a cursor as visual feedback are shaping up to be a potential alternative input for targeting in VR with button selection, as they can be at least as performant as the controller and outperform head-tracking. When assessing throughput through H1 and H2, we saw that gaze with no feedback methods outperformed the controller in throughput and matched it in movement time and subjective values. This form of gaze also outperformed the head on all those measures, with one participant stating in our post-experiment survey, "It was easy to just look at the box with my eyes and not move my head around too much." Another mentioned that gaze with no feedback was "So fast. No distractions. Seemed intuitive." We saw no significant difference between gaze with feedback methods and the controller in terms of throughput. However, a distinction was apparent with MT, where only the cursor performed poorly relative to the controller. We also saw that the MT of the cursor was similar to that of the head. When discussing the two, one participant mentioned they "Hated the head tracking option. Eye gaze cursor dot was a bit distracting (wobbled a lot)." While these are results from reasonably performant gaze on targets of 3-, 6- or 9 degrees, how do things change with degradations to replicate less performant eye tracking, especially for smaller-sized targets?

7.1. Effect of degradation

The effect of gaze accuracy on user performance is more apparent with smaller element sizes. Icon sizes in laptops

and phones can subtend an angle of 3 degrees or less based on where people sit relative to those devices (1 cm in width corresponds to approximately 1 degree of visual angle at arms length or 57 cm). How would lower gaze accuracy than available from the eye tracker used here perform on such angular sizes?

Our results for H3 show that degradations significantly affected all measures for 3-degree elements, rendering gaze input practically unusable compared to undegraded gaze inputs on elements of the same size. On the other hand, each input method, except the cursor, performed similarly for larger elements and was rated similarly to the controller in all measures except misses. As one participant puts it, "For smaller squares I needed more precision so I preferred the controller for those tasks, whereas for larger squares I could use the eye gaze quite easily."

Although we didn't inform participants which gaze blocks were degraded, we could still see clear trends in their subjective evaluations and objective performance for the smallest elements. For 3-degree elements with gaze degradations, approximately 33% of all selections resulted in misses. Here, gaze received the lowest average subjective score of 0.31 compared to 2.91 for 3-degree elements without gaze degradations. We see a 24% difference in average misses for gaze with degradations on 3-degree elements (33%) vs gaze without degradation (9%), suggesting the importance of higher quality eye tracking, for smaller UI elements. With degradations, the 7.9% miss percentages for 6- and 9-degree UI elements suggest the viability of gaze as an input even when accuracy might not be best, as long as proper designs and mitigations are in place.

On the other hand, for undegraded gaze, we did not see a significant difference in misses for 6- vs. 9-degree targets, at 3.8% and 4.0%, respectively. In this case, average miss rates actually increased, though insignificantly, for the cursor, resize, and outline methods for 9-degree targets. As the average miss rate for undegraded gaze on elements at 3, 6 and 9 degrees was 9%, 3.8% and 4% this suggests diminishing returns in additional targetable area size as a function of gaze accuracy.

7.2. Effect of gaze feedback

H4 results show that feedback significantly improved miss rates for degraded gaze input on 3-degree elements but not for degraded gaze on 6- or 9-degree elements. Feedback did not affect miss rates on undegraded gaze elements. For H5, the cursor was the only feedback type rated significantly poorly for all four in-app questions using both degraded or undegraded gaze. As we saw from H2, the cursor also performed worse than other feedback methods. One reason for poor performance with a cursor could be that, while participants visually knew precisely where the system placed their gaze, this visualization led to cursor-chase (Hatfield et al., 1996; Jacob, 1995; Majaranta & Räihä, 2002). Knowing where the cursor is also allows the user to uncomfortably and unnaturally adjust the cursor, which, on average, would add time and frustration for users to select an element (Alonso et al., 2013; Arnold & Miller, 2022; Graupner & Pannasch, 2014; Murata et al., 2012; Papelis et al., n.d.; Qian & Teather, 2017). As one user put it, "For eye gaze cursor, sometimes I would need to look at a specific section of the square if I felt the calibration was slightly off so that I could get the red cursor on the green square perfectly." Other forms of feedback could provide the required information for a correction but not be as distracting; as another participant puts it, "Some of the eye tracking felt off when looking at the target, so I would have to look at the top or bottom edge to get the right box selected." It seems like the outline and resize feedback methods provide a good compromise where the user is aware of the general area of system gaze, enabling the user to perform their intended task with less ambiguity, less unnatural eye movement, and thus at a reasonable throughput and subjective state. This opportunity would not be present with the "no feedback" case, where the user simply looks at their intended element and selects, which reduces the selection time, but at the cost of increased ambiguity, as the user guesses and trusts the eye tracking system (Alonso et al., 2013).

How valuable is feedback? One participant mentioned, "The no feedback was frustrating to not be aware of my accuracy." However, participants were unaware which blocks had degraded gaze. This raises the question of the value of feedback for targeting if gaze is highly reliable. Visual feedback can get distracting as the user now has to perceive and react to a stimulus in addition to content displayed (Helmert et al., 2008). When discussing outlining, one participant mentioned, "It was so effortless to just move my eyes without having to move anything else. Also there was no distracting red dot or resizing of boxes." Another participant mentions, "It required the least amount of exertion/effort and gave me proper feedback of what I was about to select without being too annoying." A third participant says, "Has enough feedback but still feels natural." However another participant when discussing the resize feedback methods says they find the outline method distracting, "It gives you feedback on what object you selected and it doesn't distract like the outline." This reinforces the idea that an inconspicuous signifier for targeting an element is desirable, a signifier where the user has the required information about the element targeted without having it capture their attention – and perhaps one where such feedback can be selectively ignored if users trust a highly reliable gaze system.

7.3. Low miss rates

One likely reason we see lower miss rates compared to existing work (Fernandes et al., 2023), especially for our 3-degree elements (17% vs. 9%), is that they are in a grid in the center of the FOV, rather than in ISO rings (Zhang & MacKenzie, 2007). Smaller elements in a grid are confined between a 15×15 degree square, where they are closer to the center and within areas where gaze accuracy tends to be the best (Feit et al., 2017). Furthermore, Fernandes et al. (2023) used circular elements with a 3-degree diameter, which would have a targetable area of 7.07 degrees squared,

as opposed to that of using a square with a 3-degree width, where the area would be 9 square degrees. This should allow our layout to capture additional gaze fixations that would otherwise fall on empty space.

7.3.1. Limitations and future work

This study used a generic user interface environment to have a controlled study with the potential to generalize more broadly. Certainly the varied shapes and sizes of actual targets and interfaces might vary. We do provide a foundation for understanding how element size and gaze accuracy could guide the design criteria used. Of course, there are many other manipulations that remain to be tested by future work. This is the case for the feedback methods, where more advanced designs might be useful. Other system characteristics, such as the sampling frequency of the eye tracking and the sample diversity also provide important areas of future research.

7.2. Immediate feedback

Once a participant's gaze raycast intersected with an element's collider, that element immediately provided feedback, and perhaps in a manner that can be perceived as jarring. This can be distracting when users look from one point to the next, as elements that users have no intention of selecting attract attention at a high rate. This effect is likely compounded by subpar eye-tracking (or degradations), as now visual feedback attracts attention less predictably. Implementing feedback methods to appear only on fixations would probably be less distracting and result in even higher subjective evaluations. However, users now wait for delayed feedback, which might increase the time to selection and lower throughput. Future works could predict gaze location in advance. Additionally, when looking at the cursor feedback method, we did not filter incoming eye tracking data and used a small cursor. We did not test the effect of a heavily smoothed cursor with visuals encompassing a broader area, which could reduce cursor chase and be more acceptable to a user (Choi et al., 2020; Graupner & Pannasch, 2014; Helmert et al., 2008).

7.3. Subjective values are task dependent

The value of providing feedback and maintaining low miss percentages changes based on the task being completed. The subjective evaluation of the feedback shown will strongly depend on the task performed (MacKenzie et al., 2001). For example, if feedback were not present, it would be much more damaging if the user were to target and activate the wrong element during a critical surgery than to accidentally activate a button when casually browsing the web. For our particular task, the consequence of targeting an incorrect element was relatively low – a lack of progression to the next element, which didn't necessarily have significant consequences on the participant's day. Perhaps this is one reason our subjective scores tend to be skewed upwards. If participants instead performed a task with accrued investment – such as typing a phrase with an eye-based keyboard, providing feedback but the inability to delete a mistyped phrase – would likely be much more frustrating and would alter subjective scores. The same would be true if a mistaken selection would take a long time to recover from, i.e., by launching the wrong application (Lafreniere et al., 2021).

7.4. Extended areas of application

The study findings suggest that future gaze interfaces should not provide a cursor that continuously moves relative to the gaze, but should prioritize other forms of feedback such as shading elements or objects in the scene that are likely targets. Given the interaction with not just virtual objects, but physical objects, it would be interesting to combine such techniques with physical objects that a user might want to interact with (for example, turning on a physical smart lamp through AR devices). Future studies could could explore the feasibility of such methods on frustration and fatigue as well. These also provide interesting opportunities for the application of explicit gaze input in numerous other applications. Extended Reality devices afford 3D model development and presentation or even an environment for coding and software development, and eye tracking provides a potentially more efficient and less tiring mode of interaction (Katona, 2021, 2022; Kovari & Katona, 2023; Suleman et al., 2019). Recent advances in developing serious games (for studying and supporting neurodevelopmental disorders, for example) would be most successfully by having the appropriate quality of eye tracking to allow for reduced reliability on using a cursor or feedback (Dæhlen, Heldal, Rehman, et al., 2024; Dæhlen et al., 2024). Accessibility more generally provides future research and inclusive design approaches, such as through eye tracking combined with other modalities or sensors to develop brain-computer interfaces (reddy2024towards; Katona et al., 2023), motor control interfaces (Guo et al., 2021), and robotics (Shafti et al., 2019). Likewise this would allow for unobstructed ophthalmological tests (Spiegel & Erkelens, 2024) or assessments of text legibility (Dæhlen et al., 2022).

8. Conclusion

Our study adopted current UI paradigms in VR for an ecologically valid assessment of gaze in targeting and selection. We showed that sub-1-degree error eye-tracking, with no manipulations aside from visual feedback, has the potential to match or exceed the controller – the current gold standard of targeting and selection, in numerous metrics, though not yet all as misses are still an issue. Participants generally resonate with this finding, "What a great next next gen experience," or "I want eye tracking now! After using it, I don't want to go back to head/controller!" As noted in the limitations and avenues for future research, there are a number of other conditions and stimuli that need to be tested to extend these findings. There is a lot in sight for the future of VR and human computer interaction as eye

trackers improve, our design understanding of gaze aided multimodal inputs broadens, and as we leverage algorithmic enhancements, predictive modeling, AI or contextual understanding into gaze interactions.

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